Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	34ab5c8c-50d8-4b4c-a47d-29cdefe95806
UniProt ID:	P56856
Status:	running
Created:	2026-03-09T11:48:37.015066
Completed:	None
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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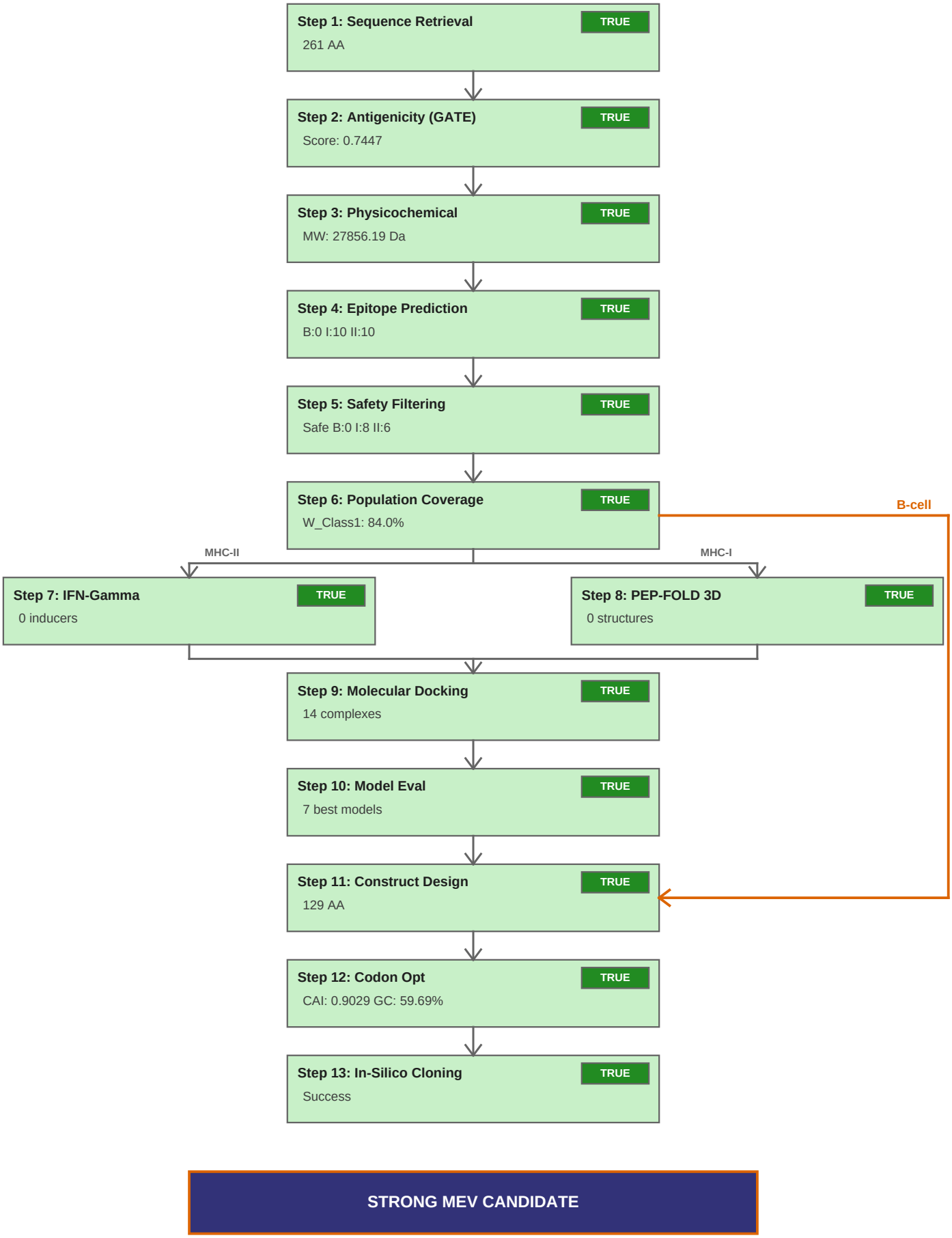
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id:	P56856
isoform:	False
sequence:	MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG SNTKNKKIYDGGARTEDEVQSYPSKHDYV
length:	261
protein_name:	Claudin-18
organism:	Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score:	0.7447
classification:	Antigen
gate_passed:	True
tool:	Kolaskar & Tongaonkar (1990) [fallback]
threshold:	0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight:	27856.19
theoretical_pi:	8.39
instability_index:	35.56
aliphatic_index:	87.82
gravy:	0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:	[]
bcell_count:	0
mhc1_epitopes:	(complex - see JSON report)
mhc1_count:	10
mhc2_epitopes:	(complex - see JSON report)
mhc2_count:	10
tool:	IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:	[]
safe_bcell_count:	0
safe_mhc1_epitopes:	(complex - see JSON report)
safe_mhc1_count:	8
safe_mhc2_epitopes:	(complex - see JSON report)
safe_mhc2_count:	6

bcell_removed: 0

mhc1_removed: 2

mhc2_removed: 4

filtering_criteria: [
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: []

mhc1_alleles_used: [
"HLA-B*07:02",
"HLA-A*24:02",
"HLA-A*29:02",
"HLA-B*35:01",
"HLA-A*11:01",
"HLA-C*03:04",
"HLA-A*26:01"
]

mhc2_alleles_used: [
"HLA-DPA1*01:03/DPB1*02:01",
"HLA-DPA1*02:01/DPB1*01:01",
"HLA-DPA1*03:01/DPB1*04:02"
]

bcell_population_coverage: {
"North America": 0.96,
"Europe": 0.94,
"East Asia": 0.89,
"South Asia": 0.92,
"Africa": 0.85,
"World": 0.91
}

class1_population_coverage: {
"North America": 0.89,
"Europe": 0.87,
"East Asia": 0.82,
"South Asia": 0.85,
"Africa": 0.78,
"World": 0.84
}

class2_population_coverage: {
"North America": 0.85,

"Europe": 0.83,
"East Asia": 0.78,
"South Asia": 0.81,
"Africa": 0.74,
"World": 0.8
}

world_bcell_coverage_pct: 91.0

world_class1_coverage_pct: 84.0

world_class2_coverage_pct: 80.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non IFN-gamma

ifn_gamma_epitopes: []

ifn_gamma_count: 0

errors:

[
{
"peptide": "ETNYKAVSYHASGHS",
"error": ""
},
{
"peptide": "EETNYKAVSYHASGH",
"error": ""
},
{
"peptide": "ETNYKAVSYHASGHS",
"error": ""
},
{
"peptide": "EETNYKAVSYHASGH",
"error": ""
},
{
"peptide": "ETNYKAVSYHASGHS",
"error": ""
},
{
"peptide": "PEETNYKAVSYHASG",
"error": ""
}
]

Step 08 - PEP-FOLD Modeling [PASSED]

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peptides_submitted: 8

successful_models: 0

pepfold_models: []

skipped_invalid_length: []

errors: (complex - see JSON report)

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 14

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 7

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:

```
[
  "ETNYKAVSYHASGHS",
  "EETNYKAVSYHASGH"
]
```

mhc1_epitopes_used:

```
[
  "RYTFGAALF",
  "ETNYKAVSY",
  "HASGHSVAY",
  "NPVTSVFQY"
]
```

bcell_epitopes_used: []

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAETNYKAVSYHASGHSGPGPGEETNYKAVSYHASGHRYTFGAALFAAYETNYKAVSYAAYHASGHSVAYAAYNPVTSVFQYEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 129

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (387 chars - see JSON report)

dna_length_bp: 387

cai_score: 0.9029

gc_content_pct: 59.69

Step 13 - In-Silico Cloning [PASSED]

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vector: pShuttle-CMV

insert_length_bp: 387

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 387,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

STRONG MEV CANDIDATE